What Makes the Best Chocolate?

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Abstract

Chocolate is a treat loved worldwide. Hence, it makes sense to study what makes chocolate so popular. For this study we analyzed a data set containing several variables relating to the production and flavor of chocolate. After performing exploratory analysis on the `rating` variable and mutating our data set, we then fit a logistic regression model to the data. From our model, and corresponding residual plots, we found that companies in Canada and France tended to produce the highest rated chocolate. Further, we found that chocolate samples with really low cocoa percent or really high cocoa percent tended to receive a lower rating. Altogether, we concluded that the best chocolate is simple in ingredient composition and produced by companies in France and Canada.

Introduction to the Problem and Question of Interest

Chocolate is a delicacy beloved by millions of people worldwide. Thus, it is relevant to research the question of what exactly makes chocolate taste so good. Specifically, stores must know what type of chocolate is in highest demand, consumers must know what types of chocolate have the best flavor, and producers must know what ingredients combine to make the best tasting chocolate. Hence, the question we will look to answer in this study, as in our title, is what makes the best chocolate?

Description of the Data

The data we will analyze contains 2,530 chocolate samples on 10 variables. The variables are as follows: `ref` (the reference number for the chocolate sample), `company_manufacturer` (company that manufactured the chocolate sample), `company_location` (country that the company is based in), `review_date` (the date the review was published), `country_of_bean_origin` (country where the bean was from), `specific_bean_origin_or_bar_name` (specific location of where the bean was from), `cocoa_percent` (percentage of cocoa in the chocolate sample), `ingredients` (number and type of ingredients in the chocolate sample), `most_memorable_characteristics` (characteristics of the chocolate sample), and `rating` (rating given by the Flavors of Cacao reviewers).

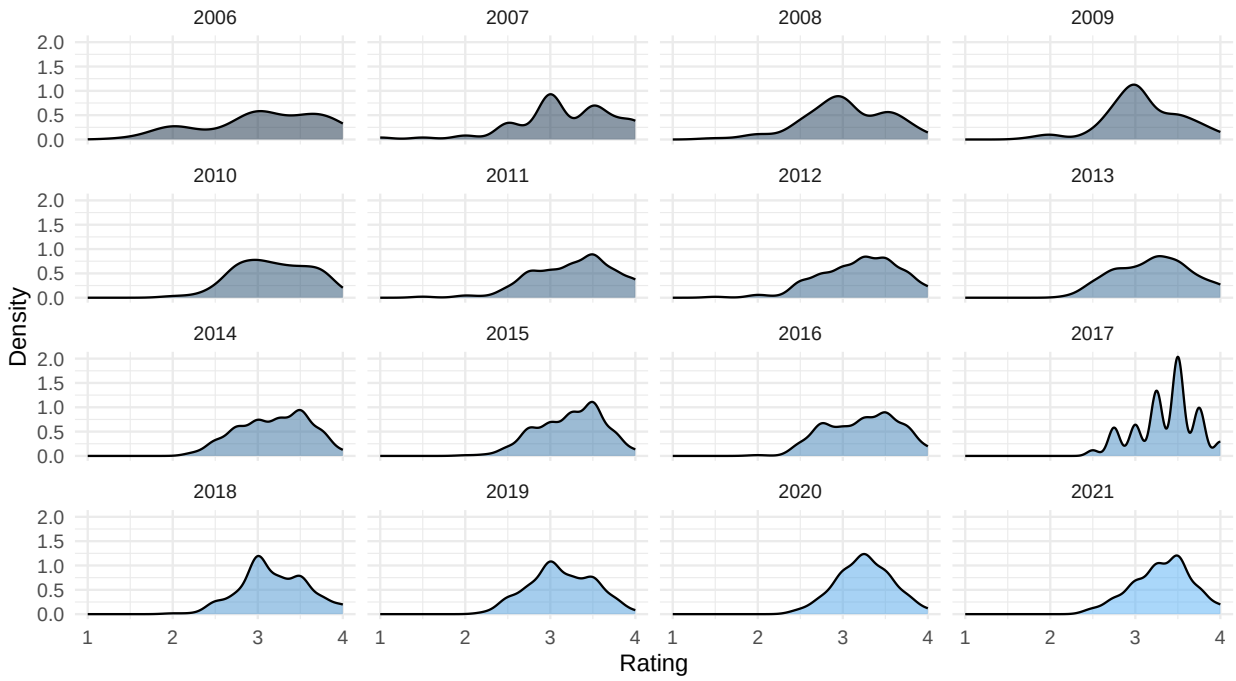
We added two, new variables to the data set called `num_ingredients` and `satisfied`.

The `num_ingredients` variable only includes the number of ingredients in the chocolate sample so we didn't have to account for all the different types of ingredients specifically used. This allows us to limit the number of dummy variables that would be necessary to build a useful model.

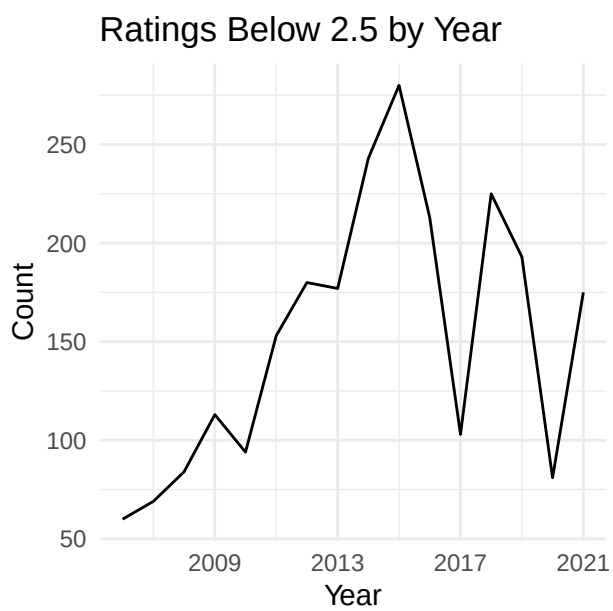
The `satisfied` variable determines whether the Flavors of Cacao reviewers gave the chocolate sample a rating greater than 3.5. We chose 3.5 as the cutoff for the definition of being satisfactory because it was representative of the 75th quantile. Therefore, we deem the best chocolate samples to be in the top 25% of all of the ones rated.

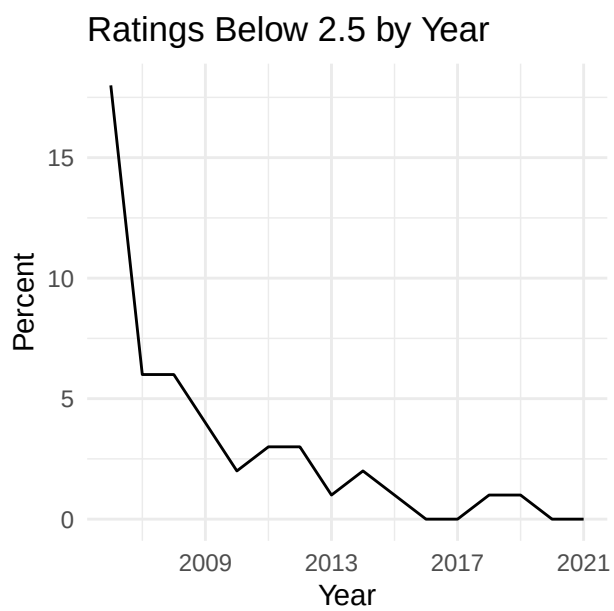
The data can be retrieved from github.com at the following link: https://github.com/rfordatascience/tidytuesday/tree/main/data/2022/2022-01-18

Analysis of the Data



The plot above shows the trend in chocolate ratings for each year given in the data set. We see that, in general, most of the ratings are in the 3 to 4 range. However, there appears to be a few ratings in the 1 to 2 range in 2006. Additionally, we can conclude based off of the plot that ratings of chocolate get better with year.





The plot, above, on the left shows the number of ratings below 2.5 by year. We can see that this number increased from 2006 to 2015 and has relatively decreased since 2015. The plot, above, on the right shows the percentage of ratings below 2.5 by year. From the plot we can see that this percentage has relatively decreased from 2006 to 2021 with the percentage, rounded to the nearest whole number, equal to 0 in 2017, 2020, and 2021.

Description of the Methods Used

As discussed earlier, our goal was to use logistic regression. Therefore, the creation of the response variable `satisfied` was necessary because it is a binary variable with possible values of 0 and 1. Chocolate samples with a rating greater than 3.5 were considered satisfactory and thus took on a value of 1. This interval was chosen by using the 75th quantile of the data. After the data was fit, we chose variables using stepwise regression to find the model with the lowest AIC. We found that the best model used the variables `cocoa_percent`, `num_ingredients`, and `company_location`. We used the top 5 countries where chocolate was made and used "other" as the reference level for company location. We also fit a model using a quadratic term for cocoa percent because the data appeared to have a quadratic relationship. However, the results were not significantly different compared to the model without the quadratic term.

In order to get the model with the lowest AIC, we started off by using a model with the number of ingredients as the only predictor variable. Then, we added cocoa percent and observed that the AIC decreased, which indicates that it is a better model. We also added company location, and that decreased the AIC as well. In addition to this, we tried adding other variables including the country of bean origin and the review date, and found that the AIC increased when those variables were added. Therefore, we concluded that the best model for the data set has the variables `cocoa_percent`, `num_ingredients`, and `company_location` (see the Appendix for our model selection progression).

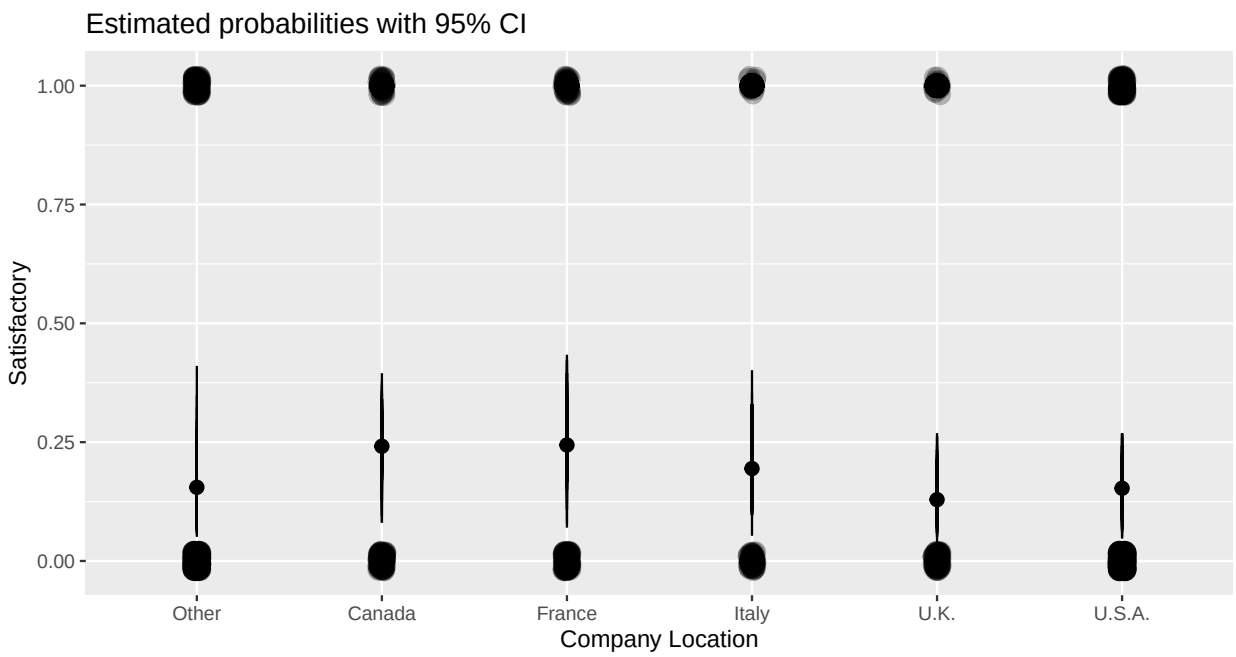
Description of the Results

```
##
## Call:
## glm(formula = satisfied ~ num_ingredients + cocoa_percent + company_location,
##     family = binomial(), data = chocolate)
##
## Coefficients:
##                        Estimate Std. Error z value Pr(>|z|)
## (Intercept)             0.44715    0.86851   0.515  0.60666
## num_ingredients        -0.08941    0.06539  -1.367  0.17157
## cocoa_percent          -0.02617    0.01122  -2.333  0.01965 *
## company_locationCanada  0.53584    0.20412   2.625  0.00866 **
## company_locationFrance  0.63030    0.21039   2.996  0.00274 **
## company_locationItaly   0.25073    0.31418   0.798  0.42485
## company_locationU.K.   -0.17757    0.28625  -0.620  0.53504
## company_locationU.S.A. -0.03950    0.13097  -0.302  0.76295
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 2197.7  on 2442  degrees of freedom
## Residual deviance: 2174.4  on 2435  degrees of freedom
## AIC: 2190.4
##
## Number of Fisher Scoring iterations: 4
```

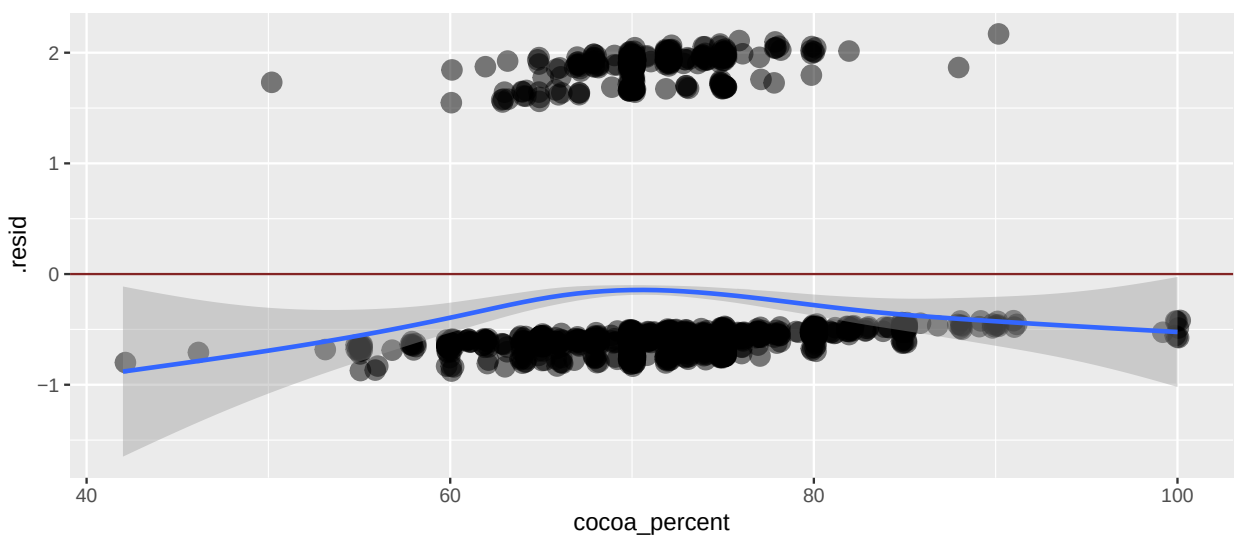
```
##            (Intercept)        num_ingredients          cocoa_percent
##              1.5638486              0.9144735              0.9741654
## company_locationCanada company_locationFrance  company_locationItaly
##              1.7088806              1.8781763              1.2849581
##   company_locationU.K. company_locationU.S.A.
##              0.8373025              0.9612683
```

According to our model, the only variables that significantly effect the estimated log odds of a satisfactory location are the cocoa percent, company location: Canada, and company location: France. A one unit increase in cocoa percent was associated with an estimated .97 multiplicative decrease in odds of satisfaction. Chocolates from Canada and France had an estimated 1.7 and 1.9 multiplicative increase in odds, respectively.

Discussion of the Results

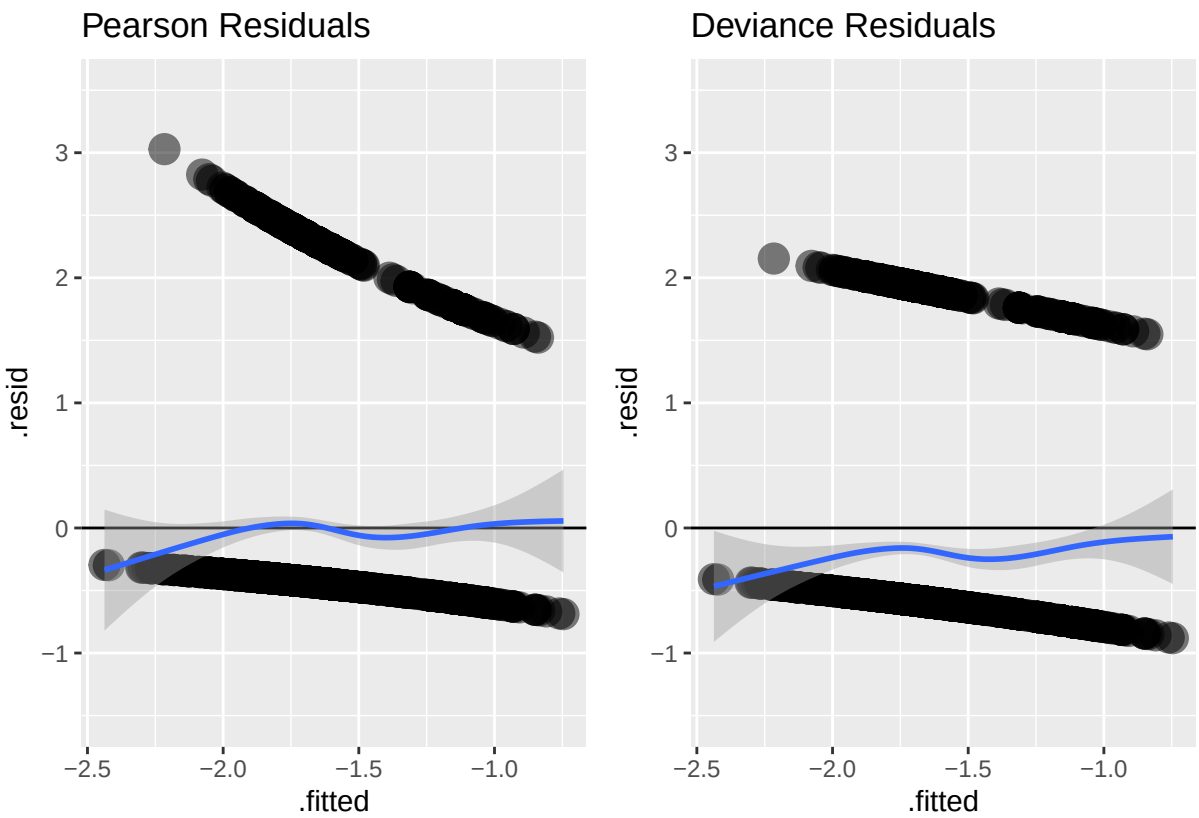


This first plot, above, shows the estimated probabilities of receiving a satisfactory rating based on our model. Canada and France have the highest point estimates at around .25. It makes sense that our point estimates are this low because only the top 25% of chocolates were satisfactory.



This second plot, above, shows the residuals plotted against cocoa percent. The residuals are approximately zero for cocoa percents between 60 and 80. This means that the estimated probability of being satisfactory is approximately the same as the actual probability. At the tails, the residuals are lower, meaning that the estimated probability is higher than the actual probability. The model most accurately estimates chocolate

samples with a medium cocoa percentage to be satisfactory. This plot looks like our data may have a parabolic relationship, but when constructing this plot using the model with a quadratic term for cocoa percent, the residuals are not significantly different (see Appendix).



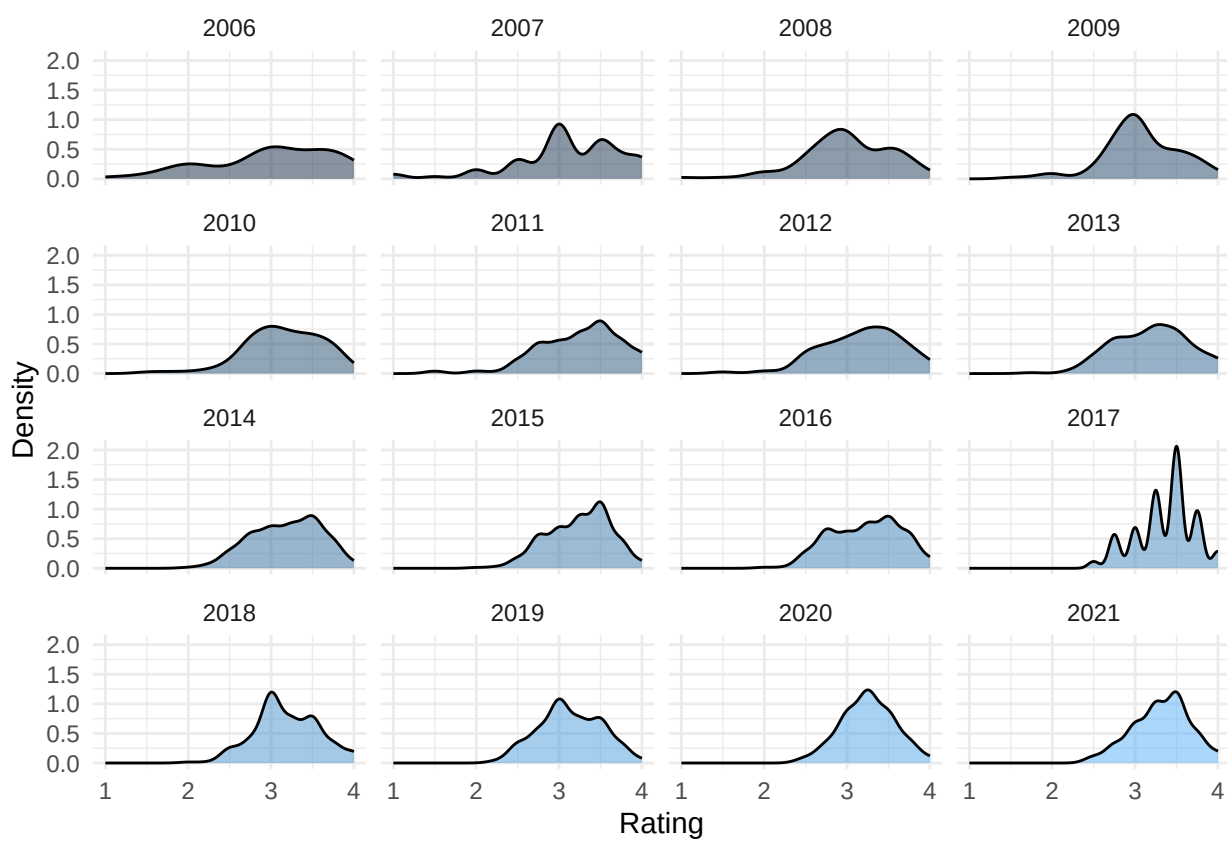
The final two residual plots, shown above, show that our residuals are approximately 0 when plotted against the fitted values, indicating that our model fit is appropriate for the data.

Conclusion

For this study, we aimed to determine what makes chocolate more likely to be rated as "satisfactory", defined as having a rating above 3.5. After exploring several models using logistic regression, we selected a final model that included cocoa percent, number of ingredients, and company location. Our results showed that as cocoa percent increases, the odds of receiving a satisfactory rating decrease slightly. Similarly, chocolates with more ingredients are less likely to be rated as satisfactory, suggesting that simpler recipes may be preferred by reviewers. Additionally, chocolates produced in Canada and France were significantly more likely to receive satisfactory ratings compared to the reference group. Overall, our model supports the idea that the best chocolate is often simpler in composition and comes from specific regions known for chocolate production.

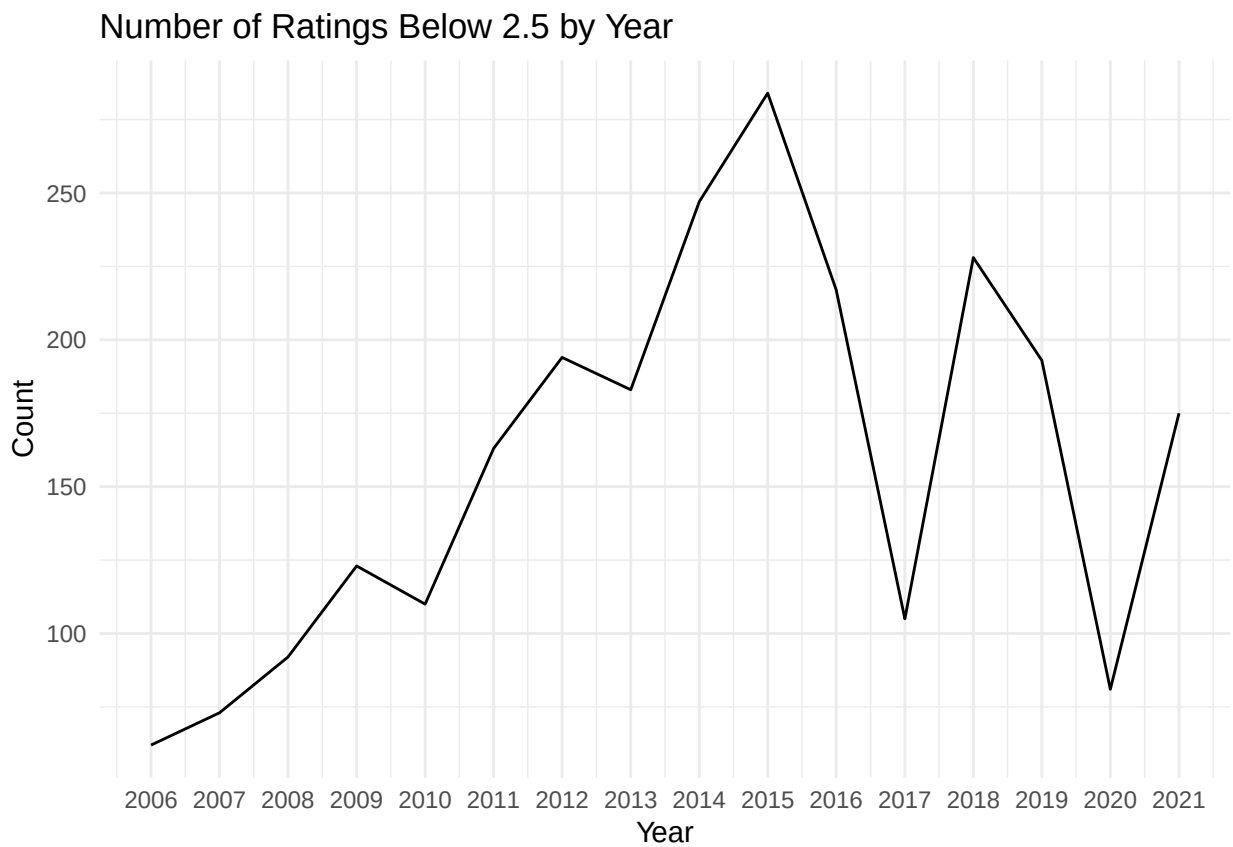
Appendix

```r
library(readr)
library(tidyverse)
library(ggplot2)
library(broom)
library(patchwork)

chocolate <- read.csv("C:/Users/bbill/Downloads/chocolate.csv")

chocolate <- chocolate %>%
  mutate(review_date = ymd(paste(review_date, 1, 1, sep="-")))

ggplot(chocolate, aes(x = rating, fill = review_date)) +
  geom_density(alpha = .5) +
  theme_minimal() +
  facet_wrap(~ as.factor(year(review_date))) +
  guides(fill = FALSE) + labs(x = 'Rating', y = 'Density')
```

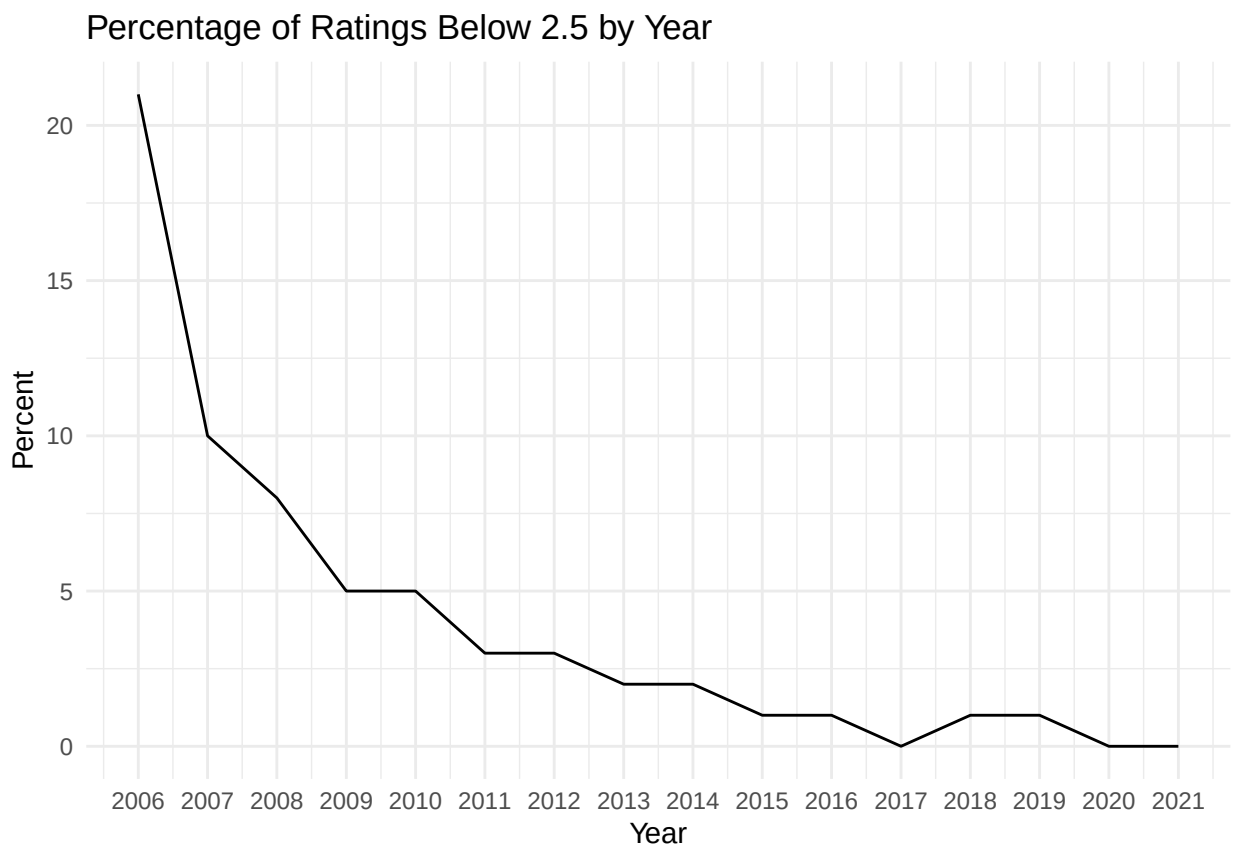



```r
low_scores <- chocolate %>%
  group_by(review_date) %>%
  summarise(less_than_2.5 = sum(rating < 2.5), count = n(),
            perc = round(less_than_2.5 / count, 2))
```

```r
low_scores %>%
  ggplot() + geom_line(aes(review_date, count)) +
  theme_minimal() +
  scale_x_date(date_breaks = '1 year', date_labels = '%Y') +
  labs(x = 'Year',
       y = 'Count',
       title = 'Number of Ratings Below 2.5 by Year')
```




```r
low_scores %>%
  ggplot() + geom_line(aes(review_date, perc * 100)) +
  theme_minimal() +
  scale_x_date(date_breaks = '1 year', date_labels = '%Y') +
  labs(x = 'Year',
       y = 'Percent',
       title = 'Percentage of Ratings Below 2.5 by Year')
```




```r
chocolate <- chocolate %>%
  mutate(
    cocoa_percent = as.numeric(str_remove(cocoa_percent, "%")),
    num_ingredients = as.numeric(str_extract(ingredients, "\\d+")),
    satisfied = if_else(rating > 3.5, 1, 0),

    company_location = fct_lump(factor(company_location), n = 5),

    company_location = fct_relevel(company_location, "Other")
  ) %>%
  drop_na(cocoa_percent, num_ingredients, company_location, satisfied)

model_logit_1 <- glm(
  satisfied ~ num_ingredients,
  data = chocolate, family = "binomial")

model_logit_3 <- glm(
  satisfied ~ num_ingredients + cocoa_percent,
  data = chocolate, family = "binomial")

model_logit_5 <- glm(
  satisfied ~ num_ingredients + cocoa_percent + company_location,
  data = chocolate, family = "binomial")

AIC(model_logit_1, model_logit_3, model_logit_5)
```

```
##               df      AIC
## model_logit_1  2 2201.601
## model_logit_3  3 2198.482
## model_logit_5  8 2190.384
```

```r
model_logit_5 <- glm(
  satisfied ~ num_ingredients + cocoa_percent + company_location,
  data = chocolate, family = "binomial")


summary(model_logit_5)
```

```
##
## Call:
## glm(formula = satisfied ~ num_ingredients + cocoa_percent + company_location,
##     family = "binomial", data = chocolate)
##
## Coefficients:
##                        Estimate Std. Error z value Pr(>|z|)
## (Intercept)             0.44715    0.86851   0.515  0.60666
## num_ingredients        -0.08941    0.06539  -1.367  0.17157
## cocoa_percent          -0.02617    0.01122  -2.333  0.01965 *
## company_locationCanada  0.53584    0.20412   2.625  0.00866 **
## company_locationFrance  0.63030    0.21039   2.996  0.00274 **
## company_locationItaly   0.25073    0.31418   0.798  0.42485
## company_locationU.K.   -0.17757    0.28625  -0.620  0.53504
## company_locationU.S.A. -0.03950    0.13097  -0.302  0.76295
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 2197.7  on 2442  degrees of freedom
## Residual deviance: 2174.4  on 2435  degrees of freedom
## AIC: 2190.4
##
## Number of Fisher Scoring iterations: 4
```

```r
exp(coef(model_logit_5))
```

```
##            (Intercept)        num_ingredients          cocoa_percent
##              1.5638486              0.9144735              0.9741654
## company_locationCanada company_locationFrance  company_locationItaly
##              1.7088806              1.8781763              1.2849581
##   company_locationU.K. company_locationU.S.A.
##              0.8373025              0.9612683
```

```r
model_quadratic <- glm(
  satisfied ~ I(cocoa_percent^2) + num_ingredients + company_location,
  data = chocolate, family = "binomial")

summary(model_quadratic)
```

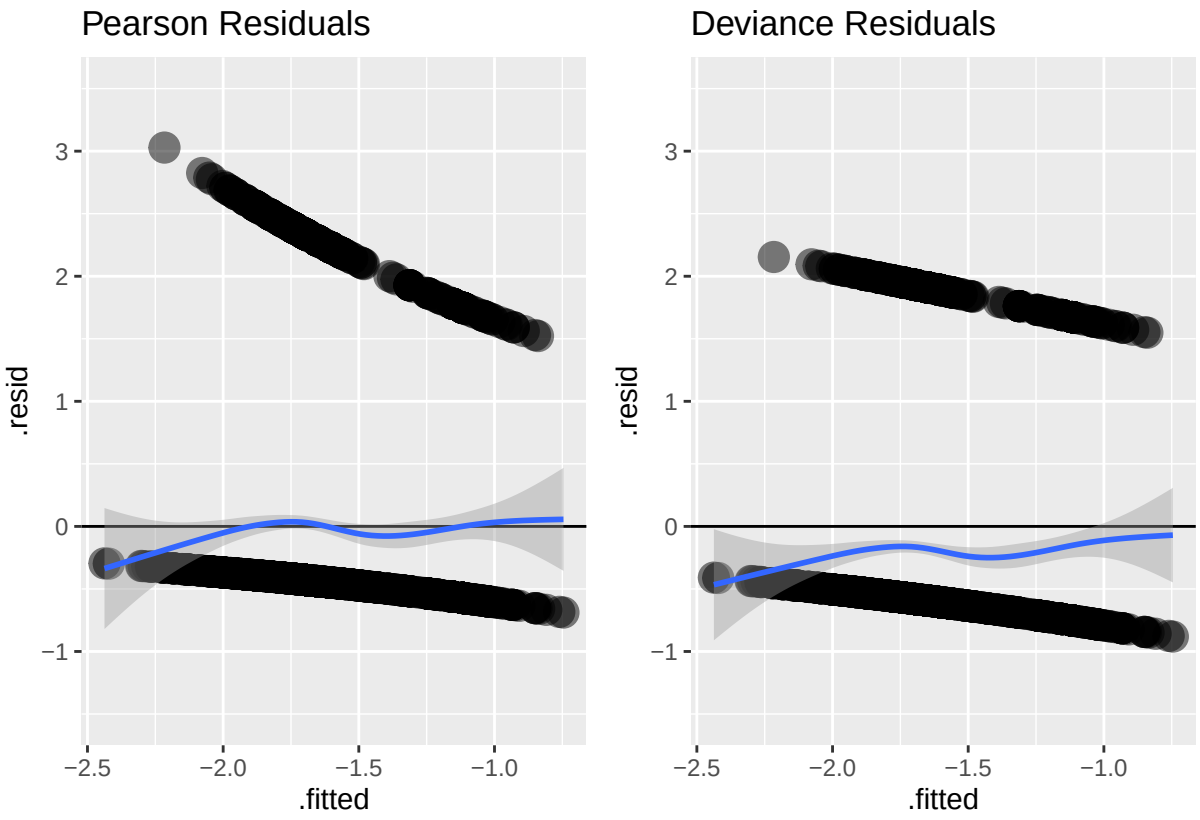
```
##
## Call:
## glm(formula = satisfied ~ I(cocoa_percent^2) + num_ingredients +
##     company_location, family = "binomial", data = chocolate)
##
## Coefficients:
##                          Estimate Std. Error z value Pr(>|z|)
## (Intercept)            -3.374e-01  4.941e-01  -0.683  0.49477
## I(cocoa_percent^2)     -2.100e-04  7.907e-05  -2.655  0.00793 **
## num_ingredients        -9.251e-02  6.532e-02  -1.416  0.15671
## company_locationCanada  5.347e-01  2.042e-01   2.618  0.00883 **
## company_locationFrance  6.335e-01  2.105e-01   3.009  0.00262 **
## company_locationItaly   2.467e-01  3.143e-01   0.785  0.43244
## company_locationU.K.   -1.736e-01  2.863e-01  -0.606  0.54433
## company_locationU.S.A. -4.169e-02  1.310e-01  -0.318  0.75033
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 2197.7  on 2442  degrees of freedom
## Residual deviance: 2172.5  on 2435  degrees of freedom
## AIC: 2188.5
##
## Number of Fisher Scoring iterations: 4
```

```r
exp(coef(model_quadratic))
```

```
##            (Intercept)     I(cocoa_percent^2)        num_ingredients
##              0.7136412              0.9997901              0.9116423
## company_locationCanada company_locationFrance  company_locationItaly
##              1.7069374              1.8841980              1.2798521
##   company_locationU.K. company_locationU.S.A.
##              0.8406535              0.9591686
```

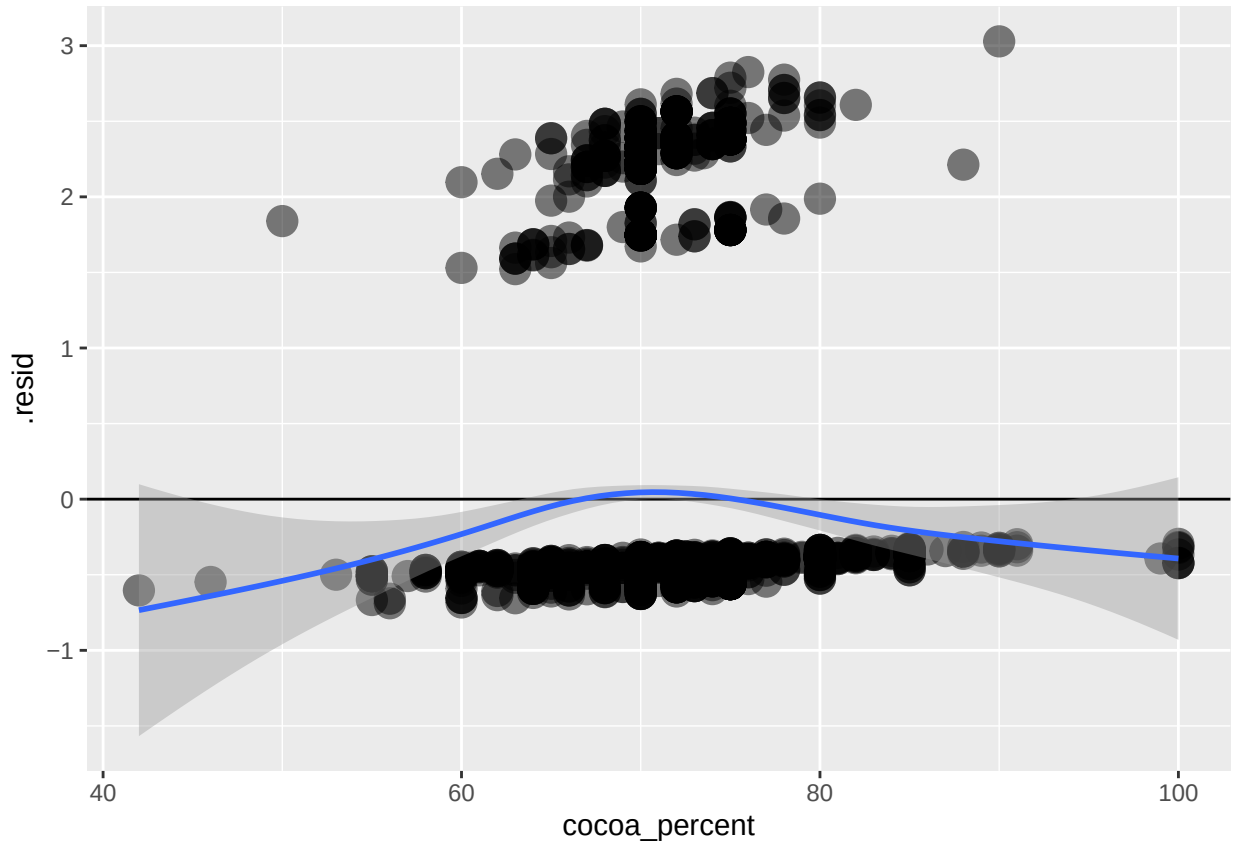
```r
p_pearson <- ggplot(augment(model_logit_5, data = chocolate,
               type.residual = "pearson"), aes(.fitted, .resid)) +
  geom_point(size = 5, alpha = 1/2) +
  geom_hline(yintercept = 0) +
  stat_smooth(span = 9/10) +
  labs(title = "Pearson Residuals")

p_deviance <- ggplot(augment(model_logit_5, data = chocolate,
               type.residual = "deviance"), aes(.fitted, .resid)) +
  geom_point(size = 5, alpha = 1/2) +
  geom_hline(yintercept = 0) +
  stat_smooth(span = 9/10) +
  labs(title = "Deviance Residuals")

print(p_pearson + ylim(-1.5, 3.5) | p_deviance + ylim(-1.5, 3.5))
```

```
## `geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'
## `geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'
```



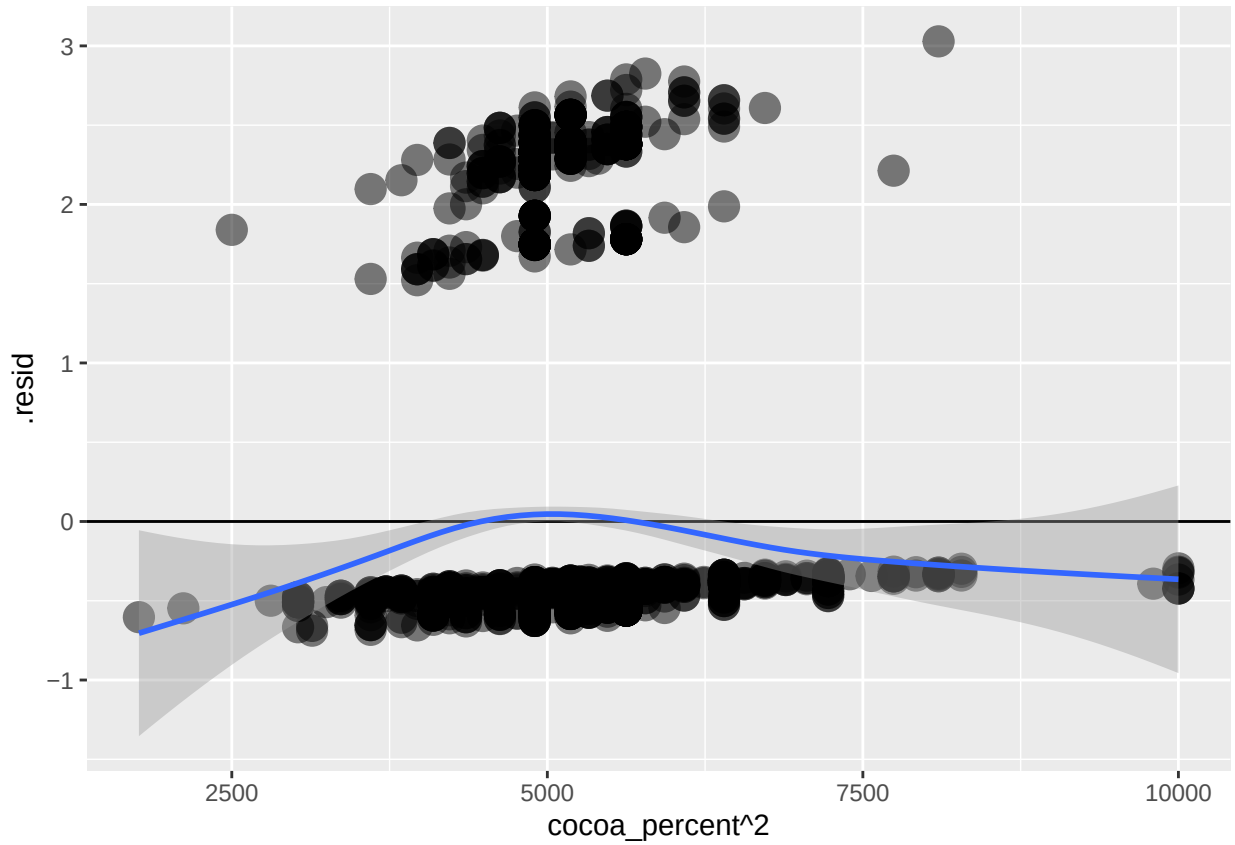

```
ggplot(augment(model_logit_5, data = chocolate,
               type.residual = "pearson"), aes(cocoa_percent, .resid)) +
  geom_point(size = 5, alpha = 1/2) +
  geom_hline(yintercept = 0) +
  stat_smooth(span = 9/10)
```

```
## `geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'
```



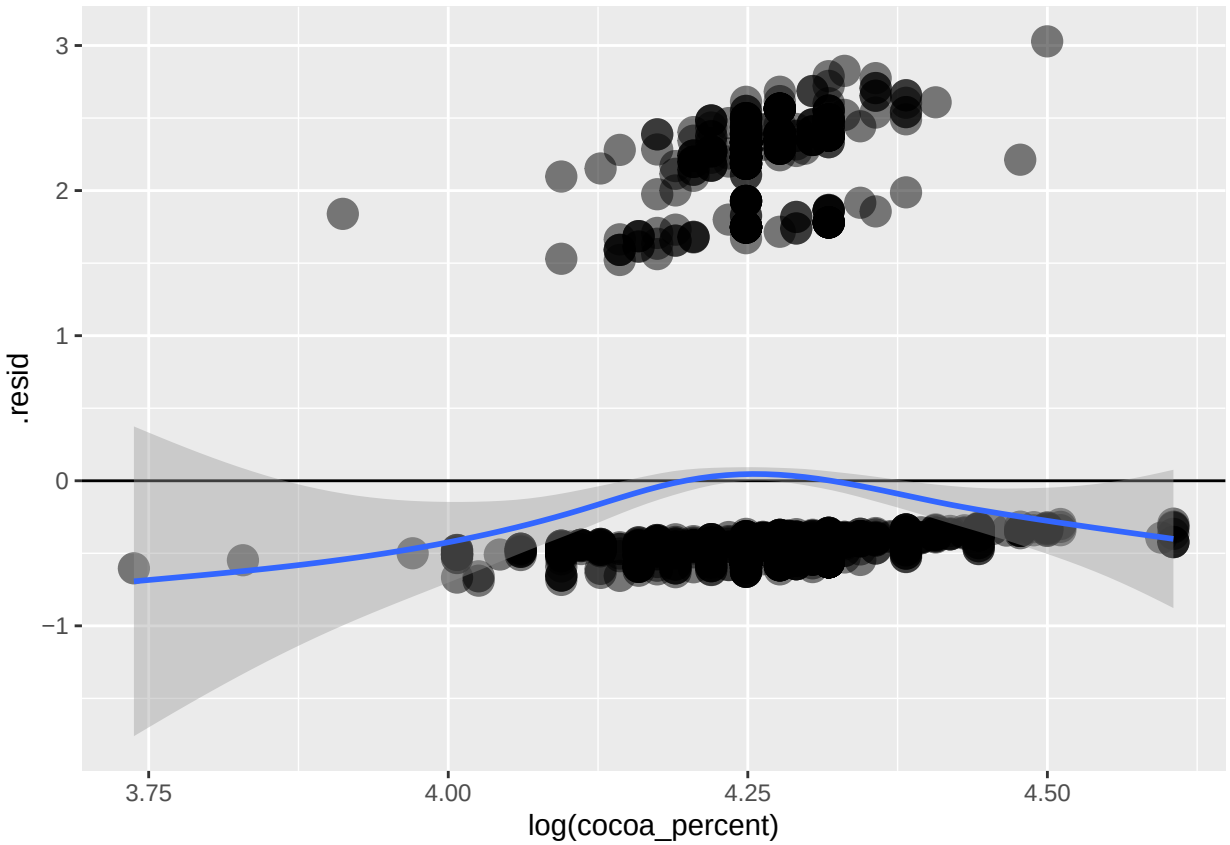

```
ggplot(augment(model_logit_5, data = chocolate,
               type.residual = "pearson"), aes(cocoa_percent^2, .resid)) +
  geom_point(size = 5, alpha = 1/2) +
  geom_hline(yintercept = 0) +
  stat_smooth(span = 9/10)
```

```
## `geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'
```




```
ggplot(augment(model_logit_5, data = chocolate,
               type.residual = "pearson"), aes(log(cocoa_percent), .resid)) +
  geom_point(size = 5, alpha = 1/2) +
  geom_hline(yintercept = 0) +
  stat_smooth(span = 9/10)
```

```
## `geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'
```


3
2
1
0
−1
.resid
3.75
4.00
4.25
4.50
log(cocoa_percent)